Benchmark run

Run

Bench id
bench-f0751dc00863

Profile
full

Seed
20240601

Suites
planted-bugs, noise-fpr, known-edge-power

Harness code hash
de45a004f42327e7e60a47c994a75acef54c61331826b8b66078
e7eea6fb57ee

Receipt hash
f0751dc0086356a2d19f8f8dc127a8f50e6f3f5e8e48d8bbc27c
e3158c217945

Overrides
s2.pnl_reconstruction=pnl_reconstruction@1

Methods used

metrics.periods_per_year=observed_density@1

s1.quality.gaps=gaps@1

s1.quality.null_rate_drift=null_rate_drift@1

s1.quality.range=range@1

s1.quality.relationship=relationship@1

s1.quality.schema=schema@1

s1.quality.timezone=timezone@1

s2.lookahead_lint=ast_rules@1

s2.pnl_reconstruction=pnl_reconstruction@1 ◆ **EXPERIMENTAL**

s2.sanity=bounds@1

s2.truncation=prefix_truncation@1

s4.walkforward=rolling_anchored@1

s5.dsr=bailey_lopezdeprado2014@1

s5.pbo=cscv@1

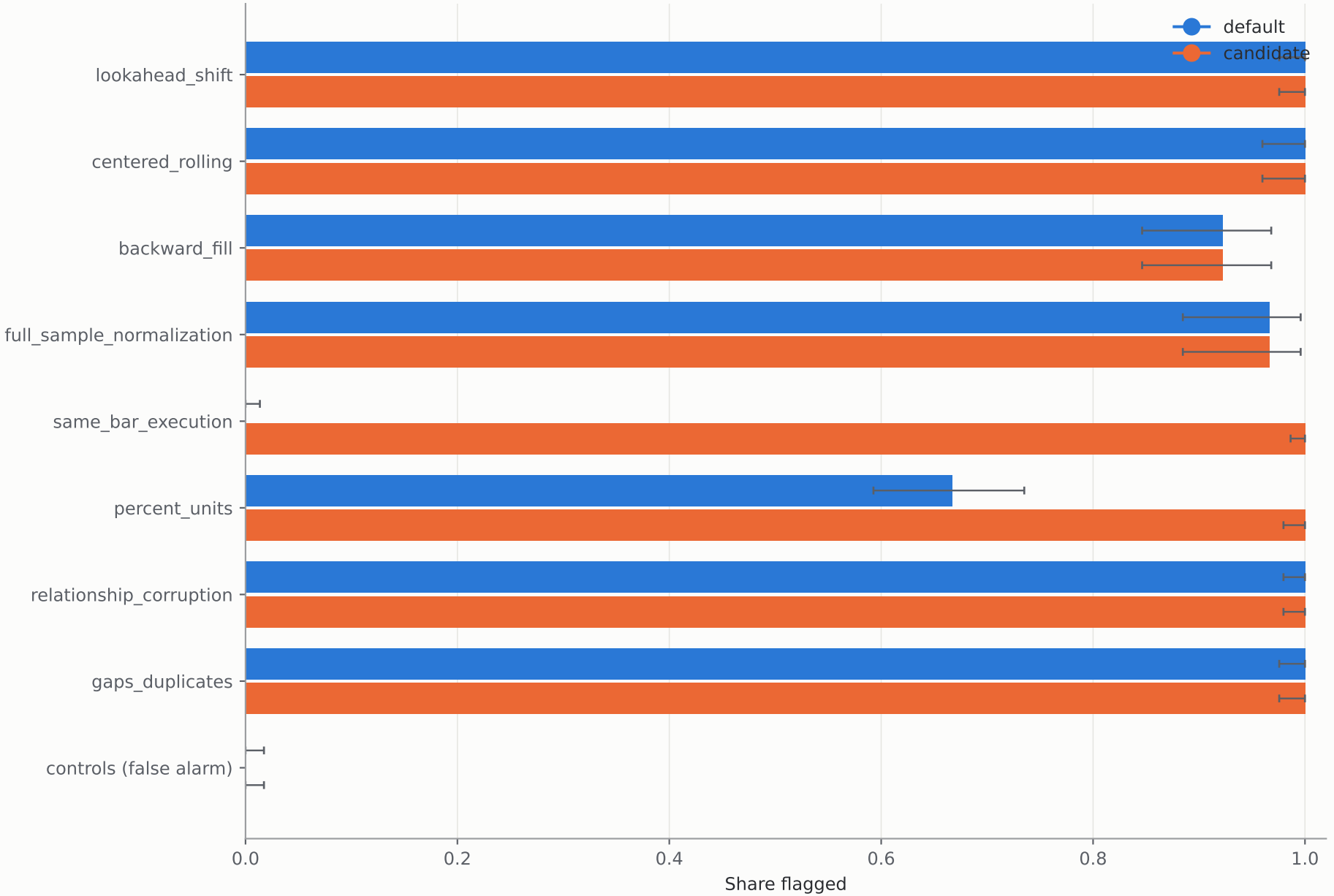
s6.bootstrap=stationary_politis_romano1994@1

EXPERIMENTAL: a non-default method version was used; it is not promoted by this report.

The default arm ran over exactly the same generated data as the candidate arm.

Synthetic/research use only, not investment advice.

Planted bugs: detection rate per bug class



Controls are clean code and data; there a lower share flagged is better.

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Planted bugs: every variant (1/3)

Bug class	Variant	Kind	Flagged (default)	Flagged (candidate)
lookahead_shift	shift_minus_1	detect	30/30	30/30
lookahead_shift	shift_minus_2	detect	30/30	30/30
lookahead_shift	shift_minus_5	detect	30/30	30/30
lookahead_shift	shift_minus_variable	detect	30/30	30/30
lookahead_shift	shift_minus_variable_k5	detect	30/30	30/30
centered_rolling	center_window_5	detect	30/30	30/30
centered_rolling	center_window_21	detect	30/30	30/30
centered_rolling	center_via_variable	detect	30/30	30/30
backward_fill	bfill	detect	30/30	30/30
backward_fill	bfill_limit	detect	30/30	30/30
backward_fill	bfill_via_getattr	detect	23/30	23/30
full_sample_normalization	price_zscore	detect	30/30	30/30
full_sample_normalization	demeaned_return	detect	28/30	28/30
same_bar_execution	same_bar_vol_0.003_raw	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.003_roll3	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.003_roll10	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.01_raw	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.01_roll3	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.01_roll10	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.02_raw	detect	0/30	30/30

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Planted bugs: every variant (2/3)

Bug class	Variant	Kind	Flagged (default)	Flagged (candidate)
same_bar_execution	same_bar_vol_0.02_roll3	detect	0/30	30/30
same_bar_execution	same_bar_vol_0.02_roll10	detect	0/30	30/30
percent_units	percent_vol_0.001_macross	detect	0/30	30/30
percent_units	percent_vol_0.001_momentum5	detect	0/30	30/30
percent_units	percent_vol_0.003_macross	detect	30/30	30/30
percent_units	percent_vol_0.003_momentum5	detect	30/30	30/30
percent_units	percent_vol_0.01_macross	detect	30/30	30/30
percent_units	percent_vol_0.01_momentum5	detect	30/30	30/30
relationship_corruption	mag_0.005_len_1	expected undetectable	0/30	0/30
relationship_corruption	mag_0.005_len_5	expected undetectable	0/30	0/30
relationship_corruption	mag_0.005_len_40	expected undetectable	0/30	0/30
relationship_corruption	mag_0.02_len_1	detect	30/30	30/30
relationship_corruption	mag_0.02_len_5	detect	30/30	30/30
relationship_corruption	mag_0.02_len_40	detect	30/30	30/30
relationship_corruption	mag_0.1_len_1	detect	30/30	30/30
relationship_corruption	mag_0.1_len_5	detect	30/30	30/30
relationship_corruption	mag_0.1_len_40	detect	30/30	30/30
gaps_duplicates	gap_1	detect	30/30	30/30
gaps_duplicates	gap_3	detect	30/30	30/30
gaps_duplicates	gap_10	detect	30/30	30/30

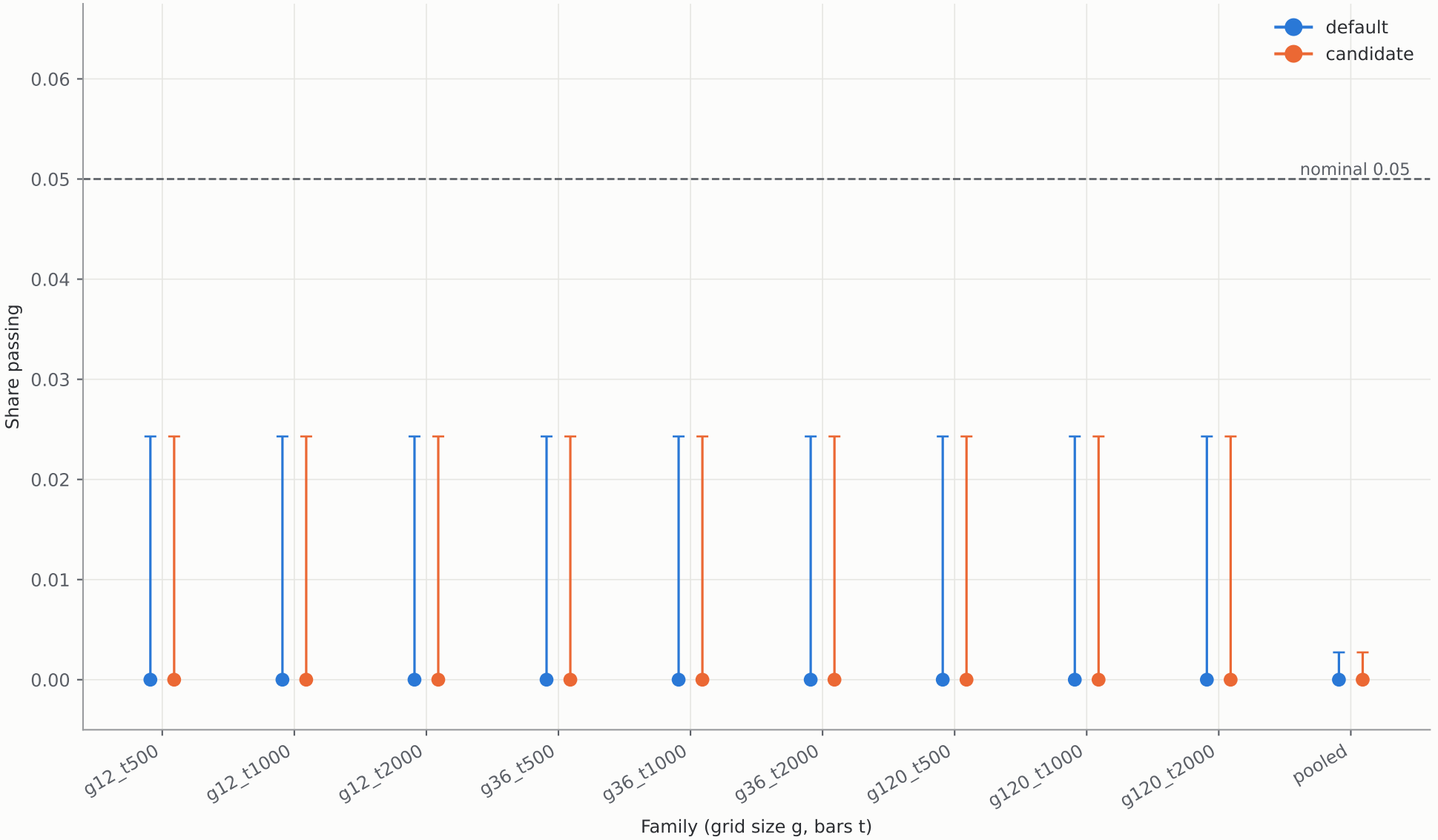
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Planted bugs: every variant (3/3)

Bug class	Variant	Kind	Flagged (default)	Flagged (candidate)
gaps_duplicates	duplicates_1	detect	30/30	30/30
gaps_duplicates	duplicates_5	detect	30/30	30/30
controls	ma_cross	false alarm	0/30	0/30
controls	return_momentum	false alarm	0/30	0/30
controls	mean_reversion	false alarm	0/30	0/30
controls	lagged_return	false alarm	0/30	0/30
controls	ewm_momentum	false alarm	0/30	0/30
controls	expanding_zscore	false alarm	0/30	0/30
controls	forward_fill	false alarm	0/30	0/30

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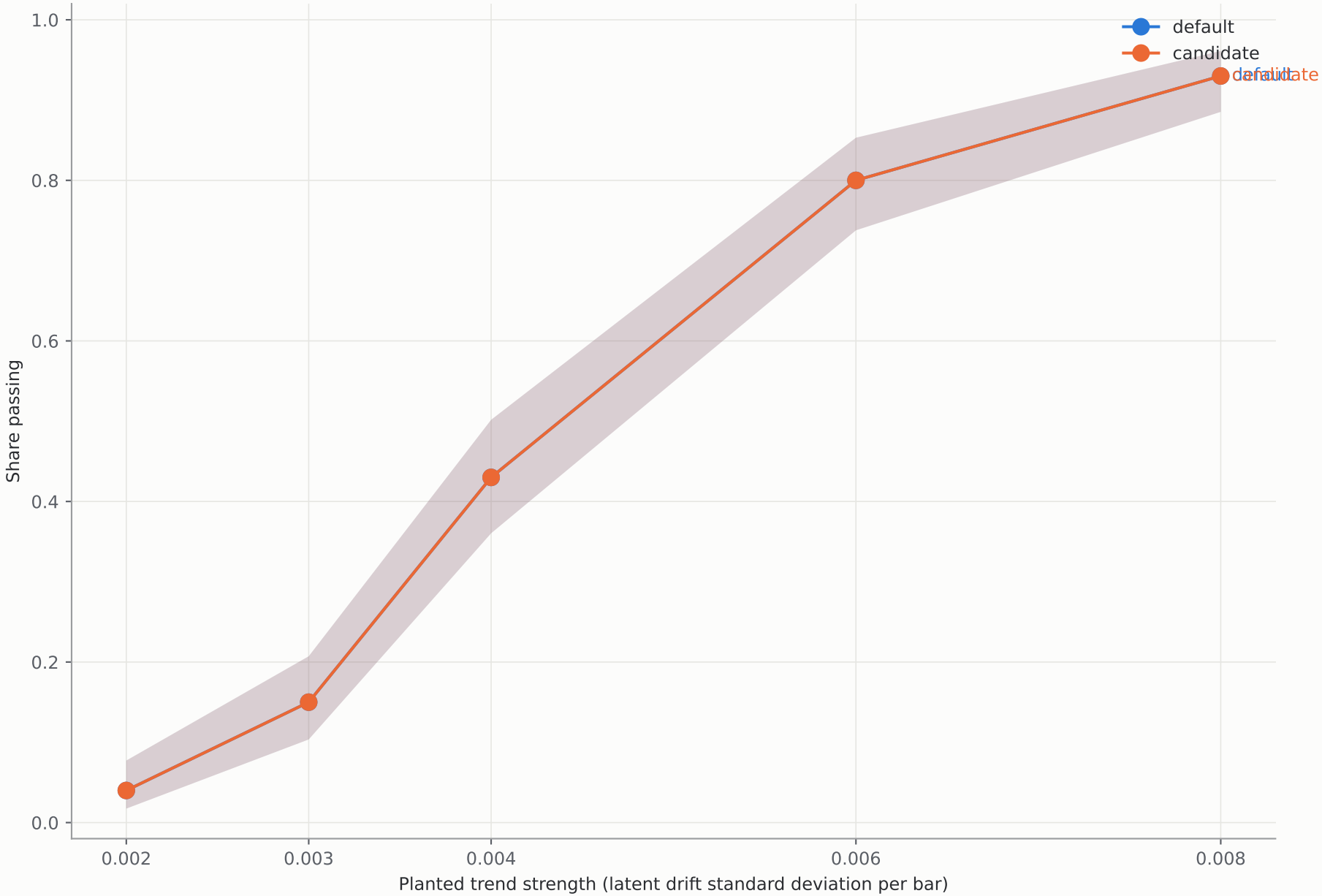
Noise false positives: share of pure-noise families passing S3-S5



Dashed line: nominal level 1 - min_dsr. Bars: exact 95% interval.

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Known-edge power: share of planted-trend datasets passing S3-S5



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Default vs candidate (gated metrics)

Metric	Default	Candidate	Delta (points)	Fisher p	Distinguishable	Better (a=default, b=candidate)
power/s0.002	8/200	8/200	0	1	no	tie
power/s0.003	30/200	30/200	0	1	no	tie
power/s0.004	86/200	86/200	0	1	no	tie
power/s0.006	160/200	160/200	0	1	no	tie
power/s0.008	186/200	186/200	0	1	no	tie
fpr/g120_t1000	0/150	0/150	0	1	no	tie
fpr/g120_t2000	0/150	0/150	0	1	no	tie
fpr/g120_t500	0/150	0/150	0	1	no	tie
fpr/g12_t1000	0/150	0/150	0	1	no	tie
fpr/g12_t2000	0/150	0/150	0	1	no	tie
fpr/g12_t500	0/150	0/150	0	1	no	tie
fpr/g36_t1000	0/150	0/150	0	1	no	tie
fpr/g36_t2000	0/150	0/150	0	1	no	tie
fpr/g36_t500	0/150	0/150	0	1	no	tie
detect/backward_fill	83/90	83/90	0	1	no	tie
detect/centered_rolling	90/90	90/90	0	1	no	tie
detect/full_sample_normalization	58/60	58/60	0	1	no	tie
detect/gaps_duplicates	150/150	150/150	0	1	no	tie
detect/lookahead_shift	150/150	150/150	0	1	no	tie
detect/percent_units	120/180	180/180	33.33	0	yes	b

both arms ran over identical generated data; Fisher's test treats them as independent samples, which is conservative

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